SPARK HACK · NVIDIA · DGX SPARK / GB10

FlowTO — Full System Architecture

A live, 100% on-device digital twin of Toronto's road + transit network.

81,669 road edges BPR + Frank-Wolfe user equilibrium ~130 tests Built 2026-05-31

FlowTO — Full System Architecture

> A live, on-device digital twin of Toronto's road + transit network. Built for the NVIDIA Spark Hack (DGX Spark / GB10). Everything runs locally: a principled traffic-assignment engine, an adaptive blast-radius recompute, an on-device Nemotron copilot, and a two-mode deck.gl/Mapbox planner's IDE. The hero scenario is FIFA WC 2026 post-match egress at BMO Field.

This document explains the entire system end to end: how data flows in, how the graph and demand are built, how the simulation engine computes equilibrium, how the blast-radius recompute keeps edits interactive, how the API serves it, and how the frontend renders it. It is written to be read top to bottom, but each section stands alone.

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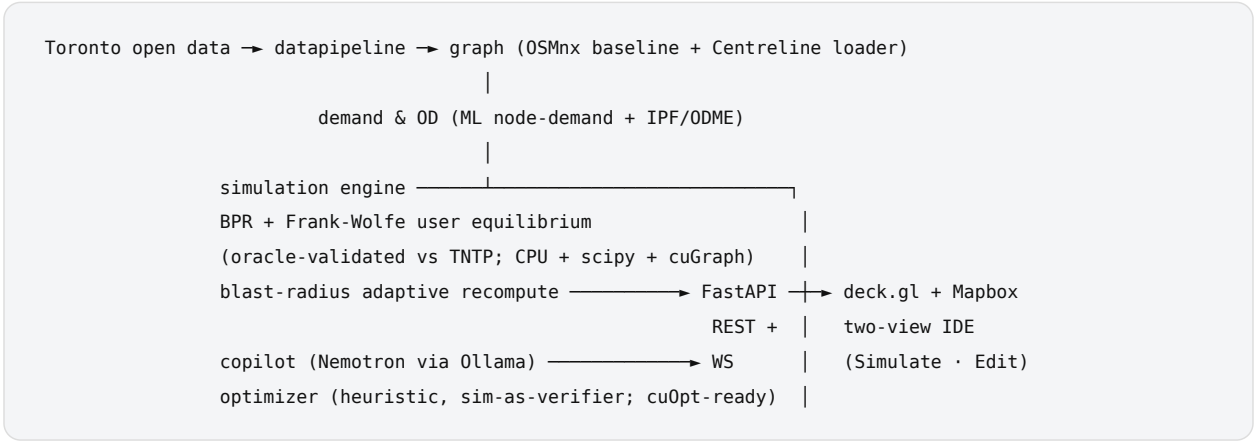
1. The 30-second mental model



The **CPU path is the demo path**. Every number you see on the map is real engine output over the actual **81,669-edge Toronto graph** — not canned data. GPU (cuGraph, cuOpt) and the LLM (Nemotron) live behind flags; they upgrade the demo but never block it. The whole stack is **deterministic**: identical inputs always produce identical outputs, which is what makes the before/after comparisons trustworthy and the tests hermetic.

Two design rules recur everywhere and are worth internalizing up front:

- **Deterministic by construction.** No wall-clock seeding of model state, stable sort orders, fixed tie-breaks in every shortest-path backend, timestamps passed in rather than read from the clock. This is why CPU/scipy/GPU backends produce bit-for-bit identical assignments.
- **Graceful degradation.** Every accelerated or external dependency has a local fallback: GPU → scipy → CPU; XGBoost → deterministic heuristic; live Nemotron → deterministic router; real GTFS → hand-authored demo routes. The demo always runs.

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2. Repository layout

```
flowTO/
├── src/torontosim/           # the real Python package (installable: `pip install -e .`)
│   ├── datapipeline/         # fetch (CKAN/GTFS/weather/restrictions) → Parquet + DuckDB + manifest
│   ├── graph/                # canonical road graph: OSMnx + Centreline, capacity calibration, schema
│   ├── model/                # ML node-demand → gravity OD → IPF/Furness → ODME → TTS/Census seed
│   ├── simulation/           # BPR + Frank-Wolfe UE; CPU/scipy/GPU backends; TNTP oracle
│   │   └── backends/         # cpu.py (heap Dijkstra), scipy_backend.py, gpu.py (cuGraph)
│   ├── blastradius/          # affected-OD detection + bounded cones + adaptive subgraph recompute
│   ├── optimizer/            # heuristic proposals scored by running the sim; cuOpt client
│   ├── copilot/              # Nemotron NL → validated tool calls; bylaw checker; local RAG
│   ├── transit/              # GTFS → route lines + scheduled vehicle trajectories
│   ├── perf/                 # timing harness + full-city vs blast-radius benchmark
│   ├── demo/                 # FIFA WC match-day surge + the three demo scenarios
│   └── api/                  # FastAPI: scenario CRUD, run/preview/compare, binary WS frames
├── src/{graph,model,simulation}/   # LEGACY pre-restructure copies (superseded by torontosim/*)
├── frontend/                 # React + Vite + TS; deck.gl + Mapbox + Zustand; landing + app
├── models/                   # demand_model.pkl (XGBoost) + gnn/ (GraphSAGE surrogate)
```

```
├── data/                    # graph JSON, raw/baked datasets, training CSVs, bench results
├── docs/                    # specs (00–12), research notes, this file
├── demo/                    # RUNBOOK.md + scenario JSON fixtures
├── scripts/                 # run_api.sh, run_frontend.sh, spark/* SSH harness
└── pyproject.toml           # package metadata + optional-dependency groups
```

> **Note on `src/graph`, `src/model`, `src/simulation`:** these are pre-"P00 restructure" copies that predate the consolidation into the `torontosim` package. The canonical, imported-everywhere code lives under `src/torontosim/`. When in doubt, follow the `torontosim` import path — that's what `pyproject.toml` packages and what the API, tests, and frontend talk to.

The package is configured in `pyproject.toml` with `package-dir = {"" = "src"}` and declares **optional-dependency groups** that map exactly onto the CPU/GPU/LLM gating: `dev`, `data`, `sim` (AequilibraE oracle), `model` (XGBoost), `gpu` (cudf/cugraph-cu13, Spark only), `gnn` (torch + PyG), `ai` (ollama + sentence-transformers + chromadb, Spark only), `api` (fastapi/uvicorn/pydantic), and `transit` (gtfs_kit).

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3. The end-to-end data flow

A single trip through the system, from raw bytes to a colored road on screen:

1. **Ingest.** `datapipeline` fetches Toronto open data (Centreline, Intersections, TMC counts, signals, bridges, neighbourhoods), GTFS feeds (TTC/GO/UP), live road restrictions, and ECCC weather. It normalizes everything to Parquet + a DuckDB catalog + a SHA-256 lineage manifest.
2. **Build the graph.** `graph` turns either the committed OSMnx JSON (default, demo-safe) or the baked Centreline Parquet into a canonical NetworkX `MultiDiGraph` with per-edge capacity, speed, free-flow time, lanes, geometry, and per-field confidence labels. TMC peaks calibrate capacity upward where observed flow exceeds the model.
3. **Predict demand.** `model.predict_node_demand` runs a weather/time-aware ML model (XGBoost, or a deterministic heuristic fallback) to estimate per-intersection throughput for a given `{hour, day_of_week, month, weather}` context.
4. **Build OD.** `model.generate_od_matrix` converts node demand into an origin→destination trip table via a time-of-day-biased gravity model, optionally balanced with IPF/Furness and grounded against observed counts with ODME.
5. **Assign.** `simulation` loads OD onto the network and solves for **user equilibrium** (BPR cost function, Frank-Wolfe/Conjugate-FW), or runs the faster k-path propagation loop. Output: per-edge `load`, `pressure`, `current_time_min`, `risk`.
6. **Edit & recompute.** A planner closes a road or boosts demand. `blastradius` detects exactly which OD pairs are affected, builds a bounded subgraph around the change, and reroutes only those — interactive instead of re-solving all of Toronto.
7. **Serve.** `api` (FastAPI) exposes scenario CRUD, run/preview/compare, the WC demo, copilot, optimizer, and transit, plus a **binary WebSocket tick stream**.
8. **Render.** The frontend uploads geometry once (keyed by edge index), then streams compact 17-byte-per-edge frames and recolors the deck.gl `PathLayer` by pressure — React never touches the per-tick data.

Everything after step 2 is keyed off a single shared, read-only baseline graph held in the API's `AppState`, so all scenarios compare against the same reference.

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4. Data pipeline

Package: `src/torontosim/datapipeline/` · **Spec:** `docs/specs/01-data-pipeline.md`

A three-verb CLI (`python -m torontosim.datapipeline {fetch,bake,verify}`) turns live City-of-Toronto open data into a verifiable offline store.

Sources & modules

MODULE	RESPONSIBILITY
`ckan.py`	City of Toronto CKAN portal client. `package_show()` , `resolve_resource(pkg, fmt, name_contains=...)` (survives resource-UUID renames), `dump_url()` , `datastore_pages()` (paginated streaming), `download_file()` (1 MB chunks, browser UA — required by City endpoints).
`gtfs.py`	Static GTFS feeds. A frozen `GtfsFeed` dataclass + `FEEDS` registry for `ttc` (via CKAN package `ttc-routes-and-schedules`), `go` and `up` (direct Metrolinx zip URLs). `fetch_feed(key, dest, date_tag=...)` .
`restrictions.py`	Live CART road-closures feed (`secure.toronto.ca/.../road_restrictions/v3`). `parse()` normalizes epoch-ms timestamps and geoPolylines into shapely `LineString` s; `fetch()` pulls live closures to simulate.
`weather.py`	ECCC hourly observations. `parse_filename()` and `fix_filenames()` repair malformed names like `weather_2023 6_00.csv` ; `categorize()` bins conditions into `clear/rain/snow/fog/storm` (with a temp ≤ −2 °C → snow fallback).
`bake.py`	Normalize raw CSV/GeoJSON → Parquet (WKT geometry, EPSG:4326 metadata) + a DuckDB catalog (one view per Parquet). Per-dataset bakers (`bake_centreline` , `bake_tmc` , …) coerce types. `verify()` checks row counts against `ROWCOUNT_FLOORS` (centreline ≥ 60k, tmc ≥ 300k, …).
`manifest.py`	Lineage. `ManifestEntry` / `Manifest` record source URL, CKAN UUID, `fetched_at` (ISO-8601, **passed in** for determinism), `sha256` , and the correct open-data license attribution per source.
`cli.py`	`fetch` (network) → `bake` (offline) → `verify` (floors). The `CKAN_DATASETS` catalog maps logical names to `(package_id, format)` . Data dir is `TS_DATA_DIR` (default `<repo>/data`).

Output layout: `data/raw/<name>.<ext>` → `data/parquet/<name>.parquet` + `data/catalog.duckdb` + `data/manifest.json` ; GTFS bakes to `data/transit/{agency}_{date}.json` .

The pipeline is split deliberately: **`fetch` is the only step that touches the network**, so `bake` / `verify` (and all tests) run hermetically offline against the raw cache.

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5. Graph

Package: `src/torontosim/graph/` · **Specs:** `docs/specs/02-graph.md` , `docs/graph-model-and-api.md`

The canonical road network is a NetworkX **`MultiDiGraph`** (parallel edges allowed, directed). Two builders feed the *same* schema so everything downstream is source-agnostic.

Representation

- **Nodes** = intersections. Attributes: `x` / `lon` , `y` / `lat` , `name` (synthesized from incident streets, e.g. "King Street West & Spadina Avenue"), `degree` .

- **Edges** = directed street segments, keyed by a deterministic `edge_id = "{u}-{v}-{k}"`. Attributes: `road_name`, `road_class` (`motorway` / `primary` / `secondary` / `tertiary` / `residential` / `service` / `default`), `length_m`, `one_way`, `speed_kmh`, `lanes`, `capacity` (veh/h, must be > 0), `base_time_min` (free-flow), `current_time_min` (congested; `"Infinity"` when closed), `status` (`open` / `closed`), `load`, `pressure` (= load/capacity), `geometry` (`[[lat, lon], …]`), `centreline_id` (TCL join key), and a **`confidence`** dict labeling each of `{lanes, speed_kmh, capacity, one_way}` as `observed | inferred | default | manual`.

Builders

- **`build_graph.py` (OSMnx, default).** `download_graph()` pulls from OpenStreetMap; `enrich_graph()` fills speed/lanes/capacity from per-class defaults in `config.py` (`VEHICLES_PER_HOUR_PER_LANE`, `DEFAULT_SPEED_KMH`, `DEFAULT_LANES`) and computes `base_time_min = length_km / speed_kmh × 60`. Capacity = lanes × per-lane throughput.
- **`centreline_loader.py` (Toronto Centreline, opt-in).** `build_centreline_graph()` joins TCL segments to the Intersections file, maps `FEATURE_CODE_DESC` to road class (`expressway → motorway`, `major arterial → primary`, …), honors `ONEWAY_DIR_CODE` (0/1/−1) to add one or both directions, and attaches bridge clearances. Preserves `centreline_id` for TMC matching.

Supporting modules

- **`schema.py`** — single source of truth. `CANONICAL_EDGE_FIELDS`, `validate_graph()` (raises `SchemaError` on missing/null/invalid fields or capacity ≤ 0), `make_edge()`, and `ensure_confidence()` to backfill legacy graphs.
- **`calibrate_capacity.py`** — `calibrate(graph, tmc_records, min_gain=1.05)` only *raises* capacity where observed TMC peak hourly flow (max 15-min interval × 4) exceeds modeled capacity by > 5 %, and re-labels that field `observed`.
- **`mutations.py`** — in-place "what-if" ops the simulator and API share: `close_edge` (status=closed, capacity=0, time=∞, stashing prior values for reversal), `reopen_edge`, `change_capacity`, `add_edge`, `remove_edge`, `close_node`.
- **`repair.py`** — `heal_oneway_arterials()` converts unreliable OSM one-ways to two-way (default `mode="all"`, or conservative `mode="named"`) so closing a segment can't silently strand trips.
- **`routing.py`** — the `edge_id → (u, v, k)` index (`build_edge_index`), nearest-node / nearest-edge spatial lookups, `find_shortest_path`, and JSON I/O (`export_graph_json` / `import_graph_json`, which converts ∞ ↔ `"Infinity"`).
- **`config.py`** — `normalise_road_class`, `haversine_m`, `base_time_min`, plus the shared `BPR_PARAMS` (α, β per road class).

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6. Demand & OD

Package: `src/torontosim/model/` · **Specs:** `docs/specs/03-demand-and-od.md`

This subsystem answers "how many vehicles want to go from where to where, right now?" in two stages: per-node demand, then origin→destination pairs.

Stage 1 — ML node demand

`features.py` defines a frozen 11-feature schema (`FEATURE_ORDER`) shared by training and inference so they can never drift:

`lat, lon, hour, day_of_week, month, is_weekend, weather_code, road_degree, distance_to_downtown, near_highway, road_class_rank`. Static per-node features are precomputed once per graph.

- **`train_demand_model.py`** trains a gradient-boosted regressor on `vehicle_count`. Backend is XGBoost (`device=cuda` on the Spark) or sklearn `HistGradientBoostingRegressor`. `train_with_sweep()` does a resumable randomized hyperparameter search. Synthetic data can be generated when real counts are unavailable. The artifact is a joblib payload `{model, feature_order, target, kind, metrics}` at `models/demand_model.pkl`.
- **`predict_node_demand.py`** loads that model and returns `{node_id: demand}` for a time context. Three-way fallback governed by `FLOWTO_DEMAND_MODEL`: **GNN** (if available) → **XGBoost** (the committed pkl) → **`HeuristicDemandModel`** — a deterministic closed-form formula (`base × class_factor × downtown_factor × highway_factor × degree_factor × rush_factor × weather_factor`) that mirrors the synthetic data generator exactly, so the demo never depends on a trained artifact loading.
- **`ingest_real_data.py`** builds the training CSVs from real TMC counts + weather, snapping each sensor to the nearest graph node via a cKD-tree and splitting train/val **by intersection** (no leakage).

Stage 2 — gravity OD + calibration

`generate_od_matrix.py` turns node demand into `[{origin, destination, trips}, …]`:

- **Gravity core:** `OD(i,j) = origin_strength(i) × destination_strength(j) / (1 + dist_km)`, with **time-of-day biasing** (AM peak pulls residential→downtown; PM peak pushes downtown→outer; weekend evenings add a nightlife pull). Restricted to the largest strongly-connected component, top ~1,500 origins/dests, trip lengths ∈ [0.4, 25] km, scaled to a nominal total (~100k).
- **`calibration="ipf"`** runs Furness biproportional fitting (`ipf.py`) to match production/attraction marginals; structural zeros stay zero.
- **`calibration="ipf_counts"`** additionally runs **ODME** (`odme.py`): all-or-nothing assign the seed OD to links, compute per-link observed/assigned ratios, and nudge each OD pair by the (damped) geometric mean of the ratios along its path — monotone error reduction, regularized toward the seed.

Supporting modules: `odme_calibrate.py` (node-throughput-based ODME against sensor counts), `timeofday.py` (TTS peak-hour factoring), `tts_seed.py` (survey/Census gravity prior exploded to nodes), `validate_past.py` (post-hoc predicted-vs-observed accuracy).

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7. Simulation engine

Package: `src/torontosim/simulation/` · **Spec:** `docs/specs/04-simulation-engine.md`

The core. Given a graph and an OD table, compute the flow on every link. There are two engines and they share one congestion model and one set of pluggable shortest-path backends.

The compact network

`network.py` converts the NetworkX `MultiDiGraph` into a `Network` dataclass: parallel NumPy arrays (`tail`, `head`, `t0`, `cap`, `alpha`, `beta`) plus a **CSR adjacency** (`indptr`, `order`) built with a stable sort for determinism. This is the form every backend operates on — cache-friendly and vectorizable.

The cost function (BPR)

`bpr.py` implements the Bureau of Public Roads volume-delay function:

```
t(x) = t0 · (1 + α·(x/c)^β),   default α = 0.15, β = 4.0,   t = ∞ if c ≤ 0
```

α/β are overridable per road class (freeways tolerate higher v/c; arterials degrade sooner). It's strictly convex, which is what guarantees a unique equilibrium.

Engine A — Frank-Wolfe user equilibrium (`equilibrium.py`)

Solves Beckmann's program `min Σ ∫₀^xᵢ tᵢ(v) dv` — the state where no driver can find a faster route (Wardrop's first principle). Three variants via the `algorithm` arg: `msa` (method of successive averages), `fw` (plain Frank-Wolfe), and `cfw` **(Conjugate Frank-Wolfe, the default)** — which uses the diagonal BPR Hessian to pick a conjugate descent direction and reaches AequilibraE-grade accuracy in far fewer iterations.

Each iteration: 1. All-or-nothing load `y` on current costs (via the selected backend). 2. Relative gap `rgap = (Σ t·x − Σ t·y) / (Σ t·x)`; stop when `rgap ≤ target`. 3. Descent direction `d = y − x` (or the conjugate blend). 4. **Exact line search** by bisection on the Beckmann derivative `Σ dᵢ·tᵢ(xᵢ + λdᵢ)`. 5. `x ← x + λd`.

Initialization is a free-flow all-or-nothing load (x = 0 is infeasible). Closed/zero-cap links get infinite cost and are never chosen. `assign_equilibrium()` wraps this and writes `load` back onto the graph.

Engine B — k-path propagation loop (`assign_paths.py` + `congestion.py`)

The original, intuitive engine and the API default. For a fixed number of iterations (4): route each origin's demand over its **top-k shortest paths** (k=3), splitting demand inversely to path travel time, then recompute congestion and repeat — so each pass routes on the *previous* pass's congested times. Path diversity comes from penalizing already-used edges (`PENALTY = 3.0`). `congestion.py` maps load → `pressure = load/cap` → `current_time_min` (either the BPR formula or a legacy piecewise multiplier) → a `risk` band, applying a weather speed factor.

Pluggable backends (`backends/`)

All implement `all_or_nothing(net, costs, od_by_origin)` and produce **bit-for-bit identical** results via shared tie-break rules (sorted origins; heap ties broken by node id; equal-cost path ties broken by lowest link index, implemented as an `index × 1e-9` epsilon in the vectorized backends):

- **`cpu.py`** — heap Dijkstra over the CSR graph. Always available; the correctness anchor.
- **`scipy_backend.py`** — `scipy.sparse.csgraph.dijkstra` over collapsed parallel edges, all origins in one C call. ~30× faster than the Python heap on the Toronto graph; used for auto-calibration.
- **`gpu.py`** — cuGraph SSSP with cached topology and a virtual super-source for multi-source cones. Spark-only; falls back to CPU on any import/runtime error.

`backends/__init__.all_or_nothing()` is the dispatcher; `available_backends()` reports what's installed.

Orchestration (`simulate_traffic.py`)

`simulate_traffic(graph, od, engine=..., congestion_model=..., backend=...)` is the front door. It optionally **auto-calibrates** total demand to a target mean pressure (0.55) — exploiting that pressure is linear in demand under fixed routing, so one assignment pass sets the scale. It captures per-iteration **frames** for animation (the equilibrium engine replays its converged flow as a load ramp so the frames build deterministically to the true answer), and returns a rich dict: `summary` (assigned trips, active edges, average pressure, high-risk/severe/closed counts, stranded trips), `frames`, the mutated `graph`, and equilibrium diagnostics (`rgap`, `converged`).

`simulate_scenario(graph, od, scenario, recompute="full"|"blast")` applies mutations then re-solves — either the whole network (always correct) or just the affected subgraph (§8). `compare_simulations()` diffs two results into a summary delta + the most-impacted edges.

Validation (`oracle.py`)

The engine's correctness is anchored on the **TNTP** academic fixtures (e.g. SiouxFalls): `oracle.py` parses TNTP net/trips/flow files, the test suite solves UE, and the Conjugate-FW link flows are checked against the *published* equilibrium — plus an optional AequilibraE cross-check (import-skipped when not installed). This is "verified, not asserted."

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8. Blast-radius

Package: `src/torontosim/blastradius/` · **Spec:** `docs/specs/05-blast-radius.md`

The problem: re-solving all of Toronto for every click is too slow to feel interactive. The insight: a local change (close one edge, throttle one corridor) only changes shortest paths in a bounded region. Blast-radius recomputes *exactly that region* and proves it equals the full recompute for the affected OD pairs.

1. **Affected-OD detection (`pathcache.py`).** Before any edit, `build_path_cache()` caches every OD pair's baseline shortest path and builds a **reverse index** `edge → set(OD)`. On a change, `PathCache.affected_ods(changed_links)` is an O(1) set lookup: exactly the OD pairs whose baseline path used a changed link.
2. **Bounded cones (`cones.py`).** `bounded_cone()` grows a cost-bounded Dijkstra **downstream** from the changed links' heads and **upstream** from their tails (default radius 8 free-flow minutes). `highway_core()` adds the endpoints of the top-capacity (90th-percentile) edges so long detours via expressways aren't missed. The union is the candidate subgraph.
3. **Reroute & widen (`recompute.py`).** `blast_assign()` reroutes only the affected OD pairs over the subgraph (`_reroute_affected`), and if boundary flow shifts by > 8 %, **adaptively widens** the radius (up to 20 min). It returns a `BlastResult` with the new per-link flow, the affected OD set, the subgraph node/link sets, and `stats` (subgraph fraction, counts) — which the frontend surfaces as "1,284 edges affected."

Both CPU (heap) and GPU (cuGraph SSSP with a virtual super-source) paths exist; GPU falls back to CPU. `tests/test_blastradius_parity.py` asserts blast output matches full recompute on the affected region.

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9. Optimizer

Package: `src/torontosim/optimizer/` · **Spec:** `docs/specs/10-optimizer.md`

Auto-proposes interventions that improve a metric, scored by **running the actual simulation** (sim-as-verifier) rather than a learned proxy — so a proposal can never look good on paper but fail in the engine.

- **`problem.py`** — `OptimizeProblem` (objective default `average_pressure`, `budget`, `max_actions`, `candidate_k`, `capacity_multiplier`, allowed `action_space`).
- **`constraints.py`** — `mask_action()` hard-rejects illegal moves: no through-capacity boosts on residential/local streets (cut-through harm), no transit-priority corridor changes, no full motorway closure (emergency access). `ACTION_COST` + `within_budget()` enforce the capital budget.
- **`search.py`** — `greedy_search()`: take the top-k congested edges, generate capacity-uplift candidates, mask the illegal ones, **score each by calling `simulate_scenario`**, sort by improvement (tie-break by edge_id), and greedily add improving actions within budget/`max_actions`. Returns `{baseline_metric, best_metric, improvement, plan, plan_cost, candidates}` and is **never worse than baseline** (empty plan if nothing helps).

- **`cuopt_client.py`** — optional NVIDIA cuOpt client for VRP sub-problems (Spark only); raises `CuOptUnavailable` and the heuristic carries the demo.

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10. Copilot

Package: `src/torontosim/copilot/` · **Spec:** `docs/specs/09-copilot.md`

Turns plain-language planner requests into **validated, schema-constrained tool calls**, read-only and preview-first by default, with hard bylaw guardrails and cited policy.

The flow (`planner.plan_intervention` → `plan.plan`):

1. **Deterministic router** handles rehearsed intents without any LLM — a blocked closure ("close Lakeshore both ways") returns a `refuse` with bylaw citations; the hero "ease gridlock near BMO Field" returns a pre-built mitigation plan.
2. **Live model (optional, Spark-gated).** Otherwise, POST to Ollama at `localhost:11434` running **`nemotron3:33b`** with `temperature=0`, `format=tool_call_json_schema()` (schema-constrained decoding), and a system prompt that mandates read-only-by-default and cite-your-policies.
3. **Schema validation.** The response is parsed into a Pydantic `ToolCall` (one of `preview_intervention`, `create_scenario`, `run_simulation`, `compare_scenarios`, `retrieve_policy`, `explain_edge`, `refuse`), with up to 3 re-asks on invalid JSON.
4. **Semantic checks.** Every `intervention.edge_id` must exist in `state.edge_index`; `constraints.check_request()` runs the deterministic **bylaw checker** (e.g. fully closing Lakeshore violates the fire-route code and TTC replacement-bus bylaw → returns `refuse` with `Violation`s); any state-changing call is forced to `requires_user_confirmation=True`.
5. **RAG grounding (`rag.py`).** A local TF-IDF bag-of-words retriever over a curated bylaw corpus (`copilot/corpus/*.md`) attaches the top policy docs to the response — no embeddings server required (the heavier `sentence-transformers`/`chromadb` stack is the Spark-only upgrade).

`explain.py` produces deterministic natural-language summaries of compare deltas ("Average network pressure eased by X…"). The `Intervention` schema is **shared** with the API (`api/schemas.py`), so the copilot emits exactly the objects the API validates and applies — plan → preview → confirm → apply.

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11. Transit overlay

Package: `src/torontosim/transit/` · **Spec:** `docs/specs/08-transit-overlay.md`

A visual overlay (not part of the assignment): scheduled transit lines and moving vehicles.

- **`gtfs_reader.py`** — stdlib reader for real GTFS zips/dirs. `route_lines()` builds per-route geometry from `shapes.txt` (falling back to ordered stop coords); `trip_trajectories()` joins `stop_times` → `stops`; `build_feed_cache()` writes `data/transit/{agency}_{date}.json`; `load_cached_feed()` reads it (falling back to the newest cached date).
- **`trajectories.py`** — `build_trajectory()` produces `{path, timestamps}` with **monotonic** seconds-since-midnight (handles GTFS > 24:00 times); `interpolate_position()` linearly interpolates a vehicle's `[lng, lat]` at any time t.
- **`routes.py`** — `ROUTE_TYPE_MODE` maps GTFS route types to modes (streetcar/subway/ bus/rail/air-rail) for coloring; ships a hand-authored `DEMO_ROUTES` (509 Harbourfront, 511 Bathurst) and `demo_trajectories()` so the overlay

always renders even without a baked feed.

The API prefers cached real feeds and falls back to the demo set.

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12. The backend API

Package: `src/torontosim/api/` · **Spec:** `docs/specs/06-backend-api.md`

FastAPI, created by `create_app(state)`. Tests inject a tiny graph; production (`serve()`) loads the full Toronto graph + baseline OD once via `_bootstrap.load_default_state()` and shares one read-only baseline.

Shared state (`store.py`)

- **`AppState`** holds the baseline `graph`, baseline `od_matrix`, the `weather` / `time_context`, a **stable `edge_ids` list** (string id → the u32 index used by binary frames) and its inverse `edge_index`, and a cached `baseline()` run (the compare reference).
- **`ScenarioStore`** is per-scenario CRUD + `run` / `preview` / `compare` orchestration over `simulate_scenario`. `edge_records(state, graph)` builds the `(idx, load, eff_speed, pressure, closed)` tuples — note effective speed is derived as `speed × base/current`.

Endpoints

ROUTE	PURPOSE
`GET /healthz`, `GET /debug/state`	health + counts
`GET /edges`	the once-uploaded edge table: `idx → edge_id + geometry/name/class`
`POST/GET/PATCH/DELETE /scenarios[/{id}]`	scenario CRUD (UUID ids, optional disk snapshots)
`POST /scenarios/{id}/run`	run with `{engine, congestion_model, backend, recompute, iterations}`; validates edge ids; returns `summary` + `blast_stats` + `rgap`
`POST /scenarios/{id}/preview`	run a hypothetical set **without** mutating stored state (asserts no scenario was added)
`GET /scenarios/{id}/compare?against=baseline`	summary delta + most-impacted edges
`GET /scenarios/{id}/records`	per-edge records of the last run (Edit-mode repaint)
`POST /copilot/plan`, `POST /copilot/explain`	copilot (501 if not installed)
`POST /optimize`	optimizer proposals (501 if not installed)
`GET /transit/{routes,trajectories}`	real cached feed → demo fallback
`GET /demo/run?scenario=baseline|wc_surge|wc_fix`	the FIFA WC demo on the real graph; deterministic → cached per scenario with a lock
`GET /jobs/{id}`	async job polling (`jobs.py` thread pool)
`WS /scenarios/{id}/stream`	binary tick frames, one per captured propagation frame, then close

The binary tick contract (`encoding.py`)

The reason the UI stays at 60 fps: geometry is uploaded **once** via `/edges`, then each edge tick is packed as a fixed 17-byte little-endian record `struct("<IfffB")` = `edge_idx:u32, load:f32, speed:f32, pressure:f32, closure:u8`, framed as `[count:u32][record × count]`. React never deserializes tick data — the typed-array store does.

Startup uses a lifespan hook that **warms only the baseline** scenario in a daemon thread (warming all three would delay the first graph response); surge and fix are computed on demand and cached.

---

13. The frontend

Package: `frontend/` · **Specs:** `docs/specs/07-frontend.md`, `docs/specs/VISUAL_TESTING.md`

React 18 + Vite + TypeScript, **two independent HTML entries** (a Vite MPA build) so the heavy map/3D code never bloats the marketing page:

- **`index.html`** → **`landing/`** — the marketing landing: Framer Motion + Lenis smooth scroll, react-three-fiber 3D scenes (CN Tower, World Cup trophy with image-based lighting), and sections (Hero, Numbers, Scenario, TwoModes, Engine, HowItWorks, CTA).
- **`app.html`** → **`main.tsx → App.tsx`** — the live planner's IDE.

The app shell

```
<TopBar/>      brand · Simulate|Edit tabs · status chip · dock toggles · theme
<ToolRail/>    Edit tools (Closure=1, Surge=2, Select=Esc)
<LeftDock/>    Saved sims (Simulate) | tool picker + scene outliner (Edit)
<MapCanvas/>   Mapbox Standard basemap + deck.gl overlay (the heart)
<BottomDock/>  timeline scrubber · playback · day-of-year picker · congestion chart
<RightDock/>   warnings ↔ details (Simulate) · Copilot chat (Nemotron)
<StatusBar/>   edges · recompute ms · subgraph edges · LLM latency · FPS · "DGX Spark · GB10"
<FirstRun/>    boot splash → "Load the twin"
```

State (Zustand)

- **`appStore.ts`** — the single UI store: theme/layout, machine state (`loaded`/`recomputing`/`status`/`planStaged`), graph + `edges`, a `pressureSeq` nonce that drives deck.gl `updateTriggers`, saved sims, Simulate selection (`selectedRoadId`), Edit state (`activeTool`, `objects`, `pendingVertices`), copilot log, timeline (`scrubberMinute`, `dayOfYear`, `playing`, `speed`), and telemetry. Actions include `loadTwin`, `applyEdits`, `placeAt`, `copilotAsk`, `applyPlan`, `selectRoad`, `recenter`, `toggleTilt`.
- **`tickStore.ts`** — module-level **typed arrays** (`load/speed/pressure/closure`) living *outside* React. `ingestFrame(buffer)` decodes a binary frame in place; `consumeDirty()` is checked once per rAF. This is the hot path — bypassing React re-renders entirely.

The map (`MapCanvas.tsx`)

Mapbox **Standard** style (time-of-day lighting via `lightPresetForMinute()` from Toronto sunrise/sunset, early 3D buildings) with a deck.gl `MapboxOverlay` interleaved in the "middle" slot. Layers: the roads `PathLayer` colored by `pressureRamp(pressure[idx])` (green→amber→red, dark-mode aware) and widened by road class; selected-road highlight; closure overlays; transit routes (mode-colored, Simulate only); the BMO Field stadium marker; pending-closure vertices; surge demand streets + directional "➤" arrows; and intervention pins. **Simulate** mode tilts to a 3D angle (pitch

52°); **Edit** mode is top-down (pitch 0°). A recompute overlay shows Demand → Assign → Pressure → Bylaw → Render progress.

Client + decoding (`api/client.ts`, `lib/decodeFrame.ts`, `api/graph.ts`)

`client.ts` wraps every REST endpoint and `connectStream(scenarioId)` opens the WebSocket; `decodeFrameInto()` mirrors the Python 17-byte format exactly. `graph.ts` reconstructs an intersection graph client-side (vertices via coordinate hashing) to power Edit-mode geometry: `corridorBetween()` (Dijkstra by road length, for closures), `streetsByDirection()` / `compassOf()` (for surge direction), and `describeSegment()` (human-readable "Road — From → To" labels, optionally upgraded via Mapbox Tilequery).

The two modes

- **Simulate** — view-only: scrub the matchday, watch egress build, click a road to chart its pressure over time, monitor warnings.
- **Edit** — place a **closure** (click two intersections → Dijkstra corridor + reverse twins → `close_edge` interventions) or a **surge** (click a street → compass radiator → `demand_surge`), which auto-calls `applyEdits()` → `POST /scenarios/{id}/run` (blast or full) → `GET …/records` → `writeRecords()` → `pressureSeq++` → recolor.

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14. GNN surrogate

Package: `models/gnn/` · **Spec:** `docs/specs/stretch/S3-gnn-surrogate.md`

A stretch upgrade to the demand path: `model.py` defines `GraphSAGEEdgePredictor` — a multi-layer GraphSAGE node encoder feeding an edge-level head that predicts congestion pressure directly (node embeddings of both endpoints + edge features + time context). `gnn_to_sim_adapter.predict_gnn_edge_state()` returns `{edge_id: {predicted_load, predicted_pressure, predicted_time_min, risk}}` and can seed the live graph. It's gated behind the `gnn` extra (torch + PyG) and selected via `FLOWTO_DEMAND_MODEL=gnn`; any failure falls back to XGBoost, then the heuristic. Training runs on the Spark (`train_on_gx10.sh`, `requirements-gx10.txt` for the aarch64/cu13 wheels).

---

15. Performance & determinism

Package: `src/torontosim/perf/` · **Spec:** `docs/specs/11-profiling-and-perf.md`

- **`timing.py`** — near-zero-overhead instrumentation: a `timer()` context manager and `@timed` decorator append `{label, ms}` to a process-global registry; `summary()` aggregates by label. Uses `perf_counter`, never sim state.
- **`bench.py`** — `benchmark()` closes the highest-pressure edge and times `simulate_scenario(recompute="full")` vs `recompute="blast"`, reporting `speedup` and `affected_subgraph_fraction`. `to_markdown()` renders the evidence table; results land in `data/bench/`.

Determinism is enforced everywhere and is load-bearing for the before/after demo and the 130-test suite: stable sorts in the CSR build, sorted origin iteration, heap ties broken by node id, equal-cost path ties broken by lowest link index (an `index × 1e-9` epsilon in the scipy/GPU backends), `Date` / `random`-free model state, and `fetched_at` timestamps passed in rather than read from the clock.

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16. CPU vs GPU vs LLM

The single most important architectural decision: **the CPU path is the demo path.** Accelerated and external dependencies are strictly additive.

CAPABILITY	LOCAL / CPU DEFAULT	ACCELERATED UPGRADE	FALLBACK ON FAILURE
Shortest paths / assignment	heap Dijkstra + scipy (`csgraph`)	cuGraph SSSP (`gpu` extra, Spark)	GPU → scipy → CPU, identical results
Demand model	committed XGBoost pkl	XGBoost `device=cuda`; GraphSAGE GNN	XGBoost → `HeuristicDemandModel` (closed form)
Copilot	deterministic router + bylaw checker + TF-IDF RAG	Nemotron-33B via Ollama; sentence-transformers/chromadb	live model → deterministic router
Optimizer	greedy + sim-as-verifier	cuOpt VRP client (Spark)	cuOpt → heuristic
Transit	hand-authored demo routes	real baked GTFS feeds	real feed → demo set

The Spark (GB10) GPU/LLM paths are validated over an SSH smoke harness (`scripts/spark/*`, `make spark-test`) against host `gx10-4f5f`, kept out of CI.

---

17. Build, test, deploy

```
make install          # py3.12/3.13 venv + pip install -e ".[dev,api,data]"
make test             # pytest -q -m "not spark"  → ~130 tests, CPU only
scripts/run_api.sh    # serve() loads the real graph + baseline → http://localhost:8000 (/docs)
cd frontend && npm install && npm run dev   # Vite proxies /api → :8000 → http://localhost:5173

# headless evidence
.venv/bin/python -m torontosim.demo.wc_surge --scenario all   # baseline → surge → fix
.venv/bin/python -m torontosim.perf.bench                     # full-city vs blast-radius
```

- **CI** (`.github/workflows/ci.yml`): Python 3.12, install `[dev,data,api]`, ruff + black, `pytest -m "not spark"`. GPU/AI/sim extras are intentionally excluded — the oracle test runs against committed TNTP fixtures and the AequilibraE cross-check is import-skipped.
- **Tests** (`tests/`, ~130): engine oracle + integration, blast-radius parity, IPF/ODME, determinism, API scenarios/demo/WS frames, copilot plan/RAG/constraints, data-pipeline bake, GTFS, packaging. Markers: `spark` (skipped in CI), `slow`, `network`.
- **Frontend tests** (`frontend/tests/`, Vitest): `decodeFrame`, `pressureRamp`, `tickStore`, `transit`, `redesign`.
- **Config knobs:** `TS_DATA_DIR`, `TS_GRAPH_JSON`, `TS_GRAPH_SOURCE` (`osmnx` | `centreline`), `TS_PARQUET_DIR`, `TS_MAX_PAIRS` (default 12000 OD pairs), `FLOWTO_DEMAND_MODEL` (`gnn` | `xgboost`), `VITE_API_TARGET`, `VITE_MAPBOX_TOKEN`.

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18. Glossary

- **BPR** — Bureau of Public Roads volume-delay function `t = t0·(1 + α(v/c)^β)`; turns link volume into congested travel time.
- **User Equilibrium (UE)** — Wardrop's first principle: a flow state where no driver can unilaterally find a faster route. Computed by Frank-Wolfe on the convex Beckmann program.
- **Frank-Wolfe / Conjugate-FW** — iterative convex solvers; CFW uses the BPR Hessian for a conjugate descent direction and converges faster.
- **All-or-nothing (AON)** — load all of an OD pair's demand onto its single current shortest path; the per-iteration subproblem of Frank-Wolfe.
- **OD matrix** — origin→destination trip table; what demand the network must carry.
- **IPF / Furness** — iterative proportional fitting; balances an OD matrix to known row/column totals.
- **ODME** — OD matrix estimation; nudges a seed OD so assigned link flows match observed counts.
- **Blast-radius** — the bounded subgraph around an edit where shortest paths actually change; recomputing only it keeps edits interactive.
- **Pressure** — `load / capacity` (v/c ratio); 0 = free-flow, ≥ 1 = gridlock; the value the map colors.
- **TNTP** — Transportation Networks for Research file format; the academic fixtures (SiouxFalls, etc.) the engine is validated against.
- **CSR** — compressed-sparse-row adjacency; the cache-friendly graph layout the backends traverse.
- **Tick frame** — the 17-byte-per-edge binary record streamed over WebSocket to recolor the map without touching React. ```
